

Significant digits : Significant digits are those digits which are used to express a no. The significant figures of a quantity are those starting from the first non-zero digit on the left to the end of last digit accurately specified on the right.

Quantity	No of S. figure
50	2
603	3
1008	4
95080	5
7.0	2
0.80	2
18.00	4
0.0003	1
0.0090	2
3.20	3
0.6046	4
2.137	4

Decimal Place 0.003 = (3D)

3.20 (1D)

0.6046 (4D)

2.137 (3D)

2.0137 (4D)

2.1370 (3D)

2.13701 (5D)

To round off a no to n significant digits discard all digits to right of the

n^{th} digit, if the $(n+1)^{\text{th}}$ digit is:

- Less than 5; Leave the n^{th} digit unchanged.
- Greater than 5; increase the n^{th} digit by 1
- Exactly 5; increase the n^{th} digit by 1 if it is odd otherwise leave it unchanged (Round off)

eg. $39.72431 \rightarrow 39.724$ (3D)

$79.07093 \rightarrow 79.071$ (3D)

$2.589951 \rightarrow 2.5900$ (2D)

$59.4865 \rightarrow 59.486$ (3D)

$47.6975 \rightarrow 47.698$ (3D).

True value X_T , Approx val = X_A

Errors:

Absolute Error: (E_A) = The absolute error of an approximate result or a measurement is the numerical difference b/w the true value of quantity and the value computed or obtained by measurement.

let $X_T \rightarrow$ True value

$X_A \rightarrow$ Approximate value

$$E_A = |X_T - X_A|$$

Relative Error: (E_R):

The R.E. = E_R is defined as the absolute error divided by the true value

$$E_R = \frac{E_A}{X_T} = \frac{|X_T - X_A|}{X_T}$$

Percentage Error (E_p): If a no is correct upto n significant figures & the first significant digit of no is k , then the relative error is less than $\frac{1}{CK \times 10^{n-1}}$.

eg. Let $N = 342.67392$
 N be correct to 8 significant digit.
 $k = 3 \rightarrow$ before decimal
 $n = 8 \rightarrow$ sig. fig

$$E_R \leq \frac{1}{3 \times 10^{8-1}} = 0.000005$$

Let $y = f(x)$

Suppose x be true value

Δx be absolute error in x

$$E_A = \Delta x$$

$$E_R = \frac{\Delta x}{x}$$

$$E_p = \frac{\Delta x}{x} \times 100\%$$

Now, $\Delta y \approx \frac{dy}{dx} \cdot \Delta x$ where Δy is absolute error in y .

Qⁿ: 3 app. of $\frac{1}{9}$ are 0.1, 0.11, 0.12 which of these 3 are best app.

so/.

$$x_T = \frac{1}{9}$$

$$|x_T - 0.10| = \left| \frac{1}{9} - \frac{1}{10} \right| = \frac{1}{90} = \frac{10}{900} \rightarrow (\text{worst})$$

$$|x_T - 0.11| = \left| \frac{1}{9} - \frac{1}{100} \right| = \frac{1}{900} \rightarrow (\text{less})$$

$$|x_T - 0.12| = \left| \frac{1}{9} - \frac{12}{100} \right| = \frac{8}{900}$$

Best app. is 0.11.

Q. Find % error in computation of $(x-y)$ for $x=12.05$
 $y=8.02$ having absolute errors
 $\Delta x = 0.005$ & $\Delta y = 0.006$

let $u = x - y$

$$\therefore x_T = 12.05 - 8.02 \quad E_A = 0.005 - 0.006 = 0.004$$

$$\therefore E_p = \frac{E_A}{x_T} \times 100 \%$$

$$= \frac{0.004}{4.03} \times 100 \%$$

$$= \frac{4}{403} \times \frac{100}{1000} \times 100$$

$$= \frac{40}{403}$$

$$= 0.099\%$$

3. If $y = 7x^7 + 3x^3$ find % error in y at $x=1$ when error of $x = 0.05$

Sol: $y = 7x^7 + 3x^3$
 $\Delta y = \frac{dy}{dx} \Delta x$

$$\frac{dy}{dx} = 49x^6 + 9x^2$$

$$y|_{x=1} = 7 \cdot 1 + 3 \cdot 1 = 10$$

$$\frac{dy}{dx} = 49x^6 + 9x^2$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 49 \cdot 1 + 9 = 58$$

$$\Delta y = 58 \times 0.05 = 2.9$$

$$\text{Percentage err} = \frac{\Delta y}{y} \times 100\%$$

$$= \frac{2.9}{10} \times 100\% = 29\%$$

$$= \frac{29}{100} \times 100 = 29\%$$

If $u = f(x_1, x_2, \dots, x_n)$ be fⁿ of several variable & Δx_i be error in x_i where $i = 1, 2, \dots, n$. Then the error Δu is given by

$$\Delta u = \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n$$

if $u = xyz^2 + x^2y^2z$ error for each of x, y, z is 0.001 at $x=2, y=4, z=5$ then compute maximum relative error in u .

$$u = xyz^2 + x^2y^2z$$

$$\Delta x = \Delta y = \Delta z = 0.001$$

$$x=2, y=4, z=5$$

$$\begin{aligned} u_T &= 2 \cdot 4 \cdot 5^2 + 2^2 \cdot 4^2 \cdot 5 \\ &= 200 + 320 \\ &= 520. \end{aligned}$$

$$\begin{aligned} \Delta u &= yz^2 \Delta x + xz^2 \Delta y + 2xy^2z \Delta z + 2x^2yz \Delta y + x^2y^2 \Delta z \\ &\quad + 2x^2yz \Delta y + x^2y^2 \Delta z \end{aligned}$$

Put values

$$= 0.774$$

$$E_R = \frac{\Delta u}{u_T} \times 100 = \frac{0.774}{520} \times 100$$

$$= 0.149\%$$

BISECTION METHOD

This method is based on the repeated application of the intermediate value theorem to obtain an approximation

to the root of the equation $f(x) = 0$.

Let $f(a) = +ve$ and $f(b) = -ve$.

Suppose $f(a)f(b) < 0$, then a root lies between a and b . The mean value $x_0 = \frac{a+b}{2}$ is taken as

the first approximate value of the root.

If $f(x_0) = 0$, then obviously $x = x_0$ is the root of the equation $f(x) = 0$.

If $f(a)f(x_0) < 0$ then, the root lies in the interval (a, x_0) . otherwise if $f(b)f(x_0) < 0$, then the root lies in the interval (x_0, b) .

we designate the new interval as $[a, b]$ whose
of length is $\frac{(b-a)}{2}$. (5)

Suppose the root lies in the interval (a, x_0) .
then, the new approximation will be.

$$x_1 = \frac{a + x_0}{2}$$

If $f(x_1)f(a) < 0$, then the root lies in the
interval (a, x_1) . otherwise the interval (x_1, x_0) .

The process length of this interval is $\frac{b-a}{2^2}$.

The process is repeated until the interval (which
contains the root) is as small as desired, say
numerically less than some given small number $\epsilon > 0$.

clearly, the interval which is reduced by one half
at each step and at the n th step the interval will be

of length $\frac{b-a}{2^n}$.

$$\therefore \frac{b-a}{2^n} < \epsilon$$

$$\Rightarrow 2^n > \frac{b-a}{\epsilon}$$

$$\Rightarrow n > \frac{\log\left(\frac{b-a}{\epsilon}\right)}{\log 2}$$

which gives the number of iterations required to
achieve the desired accuracy for the given value of ϵ .

for example,

If $|b-a| = 1$ and $\epsilon = 0.001$, we have

$$n \geq \frac{\log \frac{(b-a)}{\epsilon}}{\log 2} = \frac{\log \frac{1}{0.001}}{\log 2} = \frac{\log 10^3}{\log 2} \\ = \frac{3}{\log 2} \approx 9.9658 \\ \therefore n = 10$$

Ex: Find real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places. $\Rightarrow \epsilon = 0.001$

Solution Here $f(x) = x^3 - 2x - 5$

$$f(0) = -5$$
$$f(1) = -6$$
$$f(2) = -1$$
$$f(3) = 16$$

$\therefore$ At least one root lies between 2 and 3. $\Rightarrow b-a = 1$

let $x_0 = \frac{2+3}{2} = \frac{5}{2} = 2.5$

$$f(x_0) = f(2.5) = (2.5)^3 - 2 \times 2.5 - 5 = 5.625$$

$\therefore$ The root lies in the interval $(2, 2.5)$

$$x_1 = \frac{2+2.5}{2} = 2.25$$

$$f(x_1) = f(2.25) = 1.890625$$

$\therefore$ The root lies in the interval $(2, 2.25)$

$$x_2 = \frac{2+2.25}{2} = 2.125, \quad f(x_2) = 0.3457$$

x_i	$f(x_i)$
x_3 2.0625	-0.3513
x_4 2.09375	-0.0089
2.10938	0.1668
2.10156	0.07856
2.09766	0.03471
2.09570	0.01286
2.09473	0.00195

Ex Find a positive root of the equation $xe^x = 1$ lying in $(0, 1)$. Perform 5 iterations.

Ans 0.59275

Ex Using bisection method find the approximate root of the equation $\sin x = \frac{1}{x}$ correct to three places of decimal.

$$f(x) = x \sin x - 1$$

Ans

1.11719

Hint $f(1) = 1 \cdot \sin 1 - 1$

$$= 1 \times \sin(57.3^\circ) - 1$$

$$= -0.15849$$

$$f(1.5) = 1.5 \sin(1.5) - 1 = 0.49625$$

$$f(2) = 0.81859$$

$$b = 2$$

$$a = 1$$

$$\epsilon = 0.001$$

$$n > \frac{\log\left(\frac{b-a}{\epsilon}\right)}{\log 2}$$

$$= \frac{\log \frac{1}{0.001}}{\log 2}$$

$$= \frac{3X}{0.3010}$$

$$= 9.9666$$

Note:- In Bisection method, the interval reduces to one half at each step and at the n th step the interval will be $[a_n, b_n]$ of length $\frac{|b-a|}{2^n}$.

$$\text{Then } \frac{b-a}{2^n} < \epsilon \quad \Rightarrow \quad 2^n > \frac{b-a}{\epsilon}$$
$$\Rightarrow \quad n > \frac{\log\left(\frac{b-a}{\epsilon}\right)}{\log 2}$$

which gives us the number of iterations required to achieve the desired accuracy for given value of ϵ .

For example:- If $|b-a|=1$ and $\epsilon = 0.001$

$$\text{then } n > \frac{\log \frac{1}{0.001}}{\log 2} = \frac{\log 10^3}{\log 2} = \frac{3}{\log 2} = 9.965$$

$$\therefore n = 10$$

Convergence Analysis

Convergence of Bisection Method

$f(x) = 0$ given function

$x = \alpha$ is actual root of $f(x)$

$x_n \rightarrow$ is successive approximation of actual root $x = \alpha$

$\left\{ \begin{array}{l} x_n \text{ is said to converge to} \\ \text{actual root } x = \alpha \text{ with order } q \geq 1 \\ \text{if } |x_{n+1} - \alpha| \leq C |x_n - \alpha|^q \end{array} \right.$

$q, n \geq 0$, $C \rightarrow$ some constant

$\rightarrow$ When $q = 1$ & $0 < C < 1$ then convergence is said to be of first order.

- Order of Convergence $\rightarrow q$
- Rate of Convergence $\rightarrow C$.

$\rightarrow$ Bisection Method Always Converges:

Let α be actual root of $f(x) = 0$

$$\Rightarrow f(\alpha) = 0$$

Let x_n is successive approximation of actual root α

$$x_n = E_n + d$$

$$\Rightarrow x_{n-1} = E_{n-1} + d \quad \text{--- (1)}$$

Also

$$x_{n+1} = \frac{x_{n-1} + x_n}{2}$$

By (1)

$$\Rightarrow E_{n+1} + d = \frac{E_{n-1} + d + d + E_n}{2}$$

$$\Rightarrow E_{n+1} = \frac{E_{n-1} + E_n}{2}$$

$$\Rightarrow E_{n+1} = \frac{E_n}{2} \left[1 + \frac{E_{n-1}}{E_n} \right]$$

$\hookrightarrow$ very small
neglect it

$$\Rightarrow E_{n+1} \approx \frac{E_n}{2}$$

$$\Rightarrow E_{n+1} = \frac{1}{2} (E_n)^1 \quad \left[\begin{array}{l} E_{n+1} = A E_n^q \\ A \rightarrow \text{Rate of Convergence} \\ q \rightarrow \text{order " " } \end{array} \right]$$

$$\therefore \text{Rate of Convergence} = \frac{1}{2}$$

$$\text{Order of Convergence} = 1$$

between (2, 3), using Bisection method.

Interval is given

Solution. Let

$$f(x) = x \sin x + \cos x$$

$$f(2) = 2 \sin 2 + \cos 2 = 1.4024$$

$$f(3) = 3 \sin 3 + \cos 3 = -0.5666$$

As

$$f(2) \cdot f(3) < 0$$

$$2 < \text{Root} < 3$$

First approximate root

$$= \frac{2+3}{2} = 2.5$$

$$f(2.5) = 2.5 \sin 2.5 + \cos 2.5 = 0.6950$$

As

$$f(2.5) \cdot f(3) < 0$$

The second approximate root

$$= \frac{2.5+3}{2} = 2.75$$

$$f(2.75) = 2.75 \sin 2.75 + \cos 2.75 = 0.1253$$

As

$$f(2.75) \cdot f(3) < 0$$

453

The third approximate root $= \frac{2.75+3}{2} = 2.875$

$$f(2.875) = 2.875 \sin 2.875 + \cos 2.875 = -0.2073$$

As $f(2.875)f(2.75) < 0$

The fourth approximate root $= \frac{2.875 + 2.75}{2} = 2.8125$

$$f(2.8125) = 2.8125 \sin 2.8125 + \cos 2.8125 = -0.0374$$

As $f(2.75) \cdot f(2.8125) < 0$

The fifth approximate root $= \frac{2.8125+2.75}{2} = 2.78125$

$$f(2.78125) = 2.78125 \sin 2.78125 + \cos 2.78125 = 0.0449$$

As $f(2.78125) \cdot f(2.8125) < 0$

The sixth approximate root $= \frac{2.78125+2.8125}{2} = 2.796875$

$$f(2.796875) = 0.0040 \text{ which is nearly zero.}$$

Hence, the root of the given equation is 2.796875

An